12.55

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Question

For a matrix

$$\mathbf{M} = \begin{pmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & x \end{pmatrix} \tag{0.1}$$

The transpose of the matrix is equal to inverse of the matrix, i.e.,

$$\mathbf{M}^\top = \mathbf{M}^{-1} \tag{0.2}$$

The value of x is given by

Solution

From (0.2) it is evident that the matrix (0.1) is an orthogonal matrix. We can proceed by the fact that columns of an orthogonal matrix are pairwise perpendicular

$$\mathbf{M} = \begin{pmatrix} \mathbf{c_1} & \mathbf{c_2} \end{pmatrix} \tag{0.3}$$

where

$$\mathbf{c_1} = \begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \end{pmatrix}, \mathbf{c_2} = \begin{pmatrix} \frac{3}{5} \\ x \end{pmatrix} \tag{0.4}$$

Solution

By using (0.2):

$$\mathbf{c_1}^\top \mathbf{c_2} = 0 \tag{0.5}$$

$$\implies \begin{pmatrix} \frac{4}{5} & \frac{3}{5} \end{pmatrix} \begin{pmatrix} \frac{3}{5} \\ x \end{pmatrix} = 0 \tag{0.6}$$

$$\implies \frac{12}{25} + \frac{3x}{5} = 0 \tag{0.7}$$

$$\implies x = -\frac{4}{5} \tag{0.8}$$